

# STA 2101: Statistics & Probability

## Lecture 08: Exercise for Chebyshev

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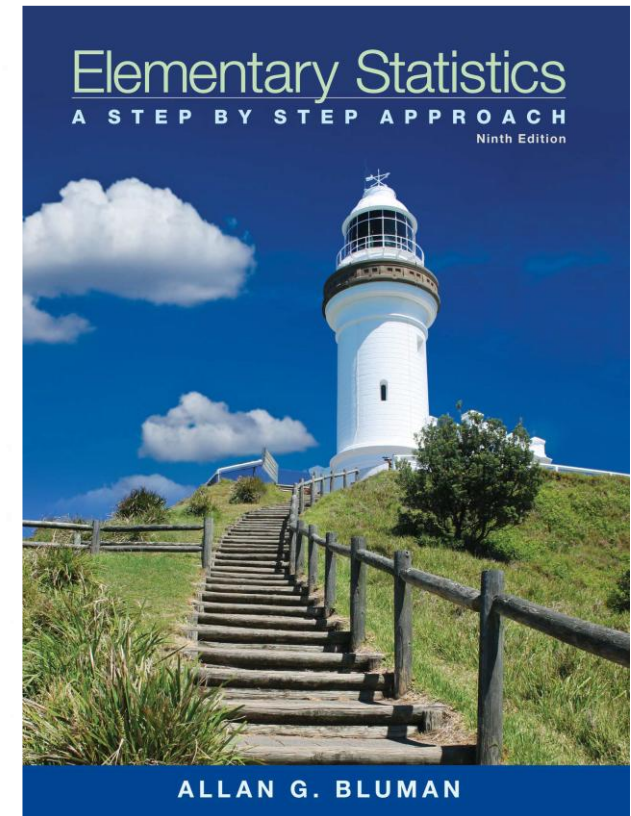
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Class Code  
Sec 1 - es2uztok  
Sec 5 - ayvzsvgm

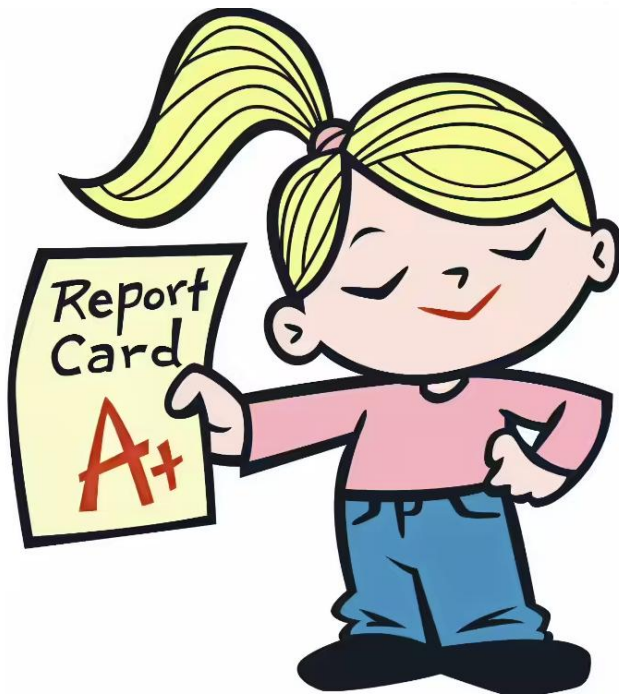
Spring 2026

# Reference Book

- **Text Book(s):**
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



# Mark Distribution



| Assessments                        | %   |
|------------------------------------|-----|
| Mid Term Examination               | 25  |
| Final Examination                  | 40  |
| Attendance and Class Participation | 5   |
| CT                                 | 5   |
| Project                            | 25  |
| Total                              | 100 |

# Today's Topics

- Cvar
- Chebyshev
- Exercises

## Coefficient of Variation

The **coefficient of variation**, denoted by CVar, is the standard deviation divided by the mean. The result is expressed as a percentage.

**For samples,**

$$\text{CVar} = \frac{s}{\bar{X}} \cdot 100$$

**For populations,**

$$\text{CVar} = \frac{\sigma}{\mu} \cdot 100$$

### EXAMPLE 3-23 Sales of Automobiles

The mean of the number of sales of cars over a 3-month period is 87, and the standard deviation is 5. The mean of the commissions is \$5225, and the standard deviation is \$773. Compare the variations of the two.

#### SOLUTION

The coefficients of variation are

$$\text{CVar} = \frac{s}{\bar{X}} = \frac{5}{87} \cdot 100 = 5.7\% \quad \text{sales}$$

$$\text{CVar} = \frac{773}{5225} \cdot 100 = 14.8\% \quad \text{commissions}$$

Since the coefficient of variation is larger for commissions, the commissions are more variable than the sales.



## Range Rule of Thumb

The range can be used to approximate the standard deviation. The approximation is called the **range rule of thumb**.

### The Range Rule of Thumb

A rough estimate of the standard deviation is

$$s \approx \frac{\text{range}}{4}$$

## Chebyshev's Theorem

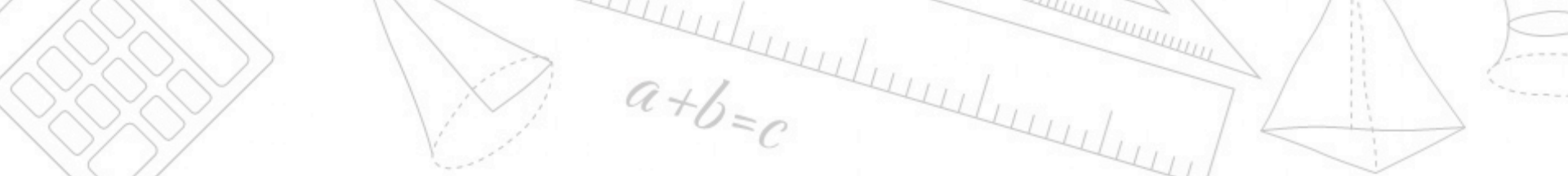
**Chebyshev's theorem** The proportion of values from a data set that will fall within  $k$  standard deviations of the mean will be at least  $1 - 1/k^2$ , where  $k$  is a number greater than 1 ( $k$  is not necessarily an integer).

This theorem states that at least three-fourths, or 75%, of the data values will fall within 2 standard deviations of the mean of the data set. This result is found by substituting  $k = 2$  in the expression

$$1 - \frac{1}{k^2} \quad \text{or} \quad 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} = 75\%$$

Furthermore, the theorem states that at least eight-ninths, or 88.89%, of the data values will fall within 3 standard deviations of the mean. This result is found by letting  $k = 3$  and substituting in the expression.

$$1 - \frac{1}{k^2} \quad \text{or} \quad 1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} = 88.89\%$$



### EXAMPLE 3–25 Prices of Homes

The mean price of houses in a certain neighborhood is \$50,000, and the standard deviation is \$10,000. Find the price range for which at least 75% of the houses will sell.

#### **SOLUTION**

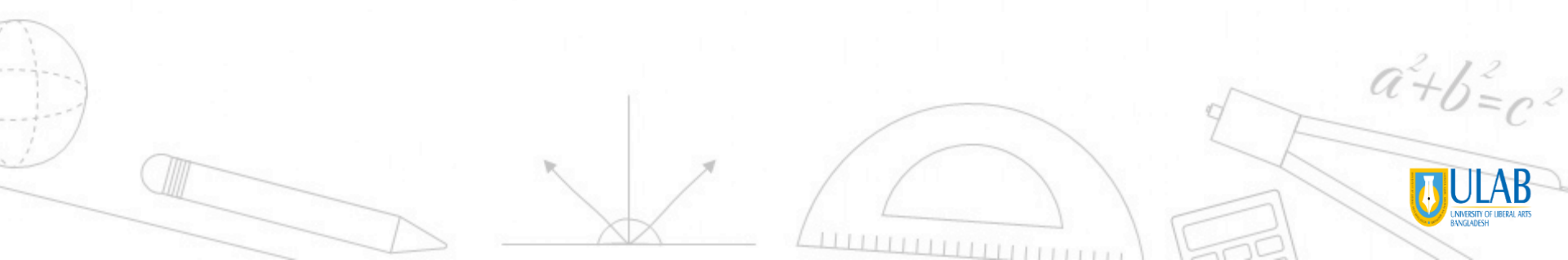
Chebyshev's theorem states that three-fourths, or 75%, of the data values will fall within 2 standard deviations of the mean. Thus,

$$\$50,000 + 2(\$10,000) = \$50,000 + \$20,000 = \$70,000$$

and

$$\$50,000 - 2(\$10,000) = \$50,000 - \$20,000 = \$30,000$$

Hence, at least 75% of all homes sold in the area will have a price range from \$30,000 to \$70,000.


$$a^2 + b^2 = c^2$$

### EXAMPLE 3–26 Travel Allowances

A survey of local companies found that the mean amount of travel allowance for couriers was \$0.25 per mile. The standard deviation was \$0.02. Using Chebyshev's theorem, find the minimum percentage of the data values that will fall between \$0.20 and \$0.30.

#### SOLUTION

**Step 1** Subtract the mean from the larger value.

$$\$0.30 - \$0.25 = \$0.05$$

**Step 2** Divide the difference by the standard deviation to get  $k$ .

$$k = \frac{0.05}{0.02} = 2.5$$

**Step 3** Use Chebyshev's theorem to find the percentage.

$$1 - \frac{1}{k^2} = 1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 1 - 0.16 = 0.84 \quad \text{or} \quad 84\%$$

Hence, at least 84% of the data values will fall between \$0.20 and \$0.30.



# Exercises

- 7. Unemployment Benefits** The average weekly unemployment benefits (in dollars) for a random selection of states are listed below. Calculate the range, variance, and standard deviation for the data.

239   214   327   416   321   289   209   356  
190   272   252   272   310   276   251

Source: *World Almanac 2012*.

- 19. Cost per Load of Laundry Detergents** The costs per load (in cents) of 35 laundry detergents tested by a consumer organization are shown here. Find the variance and standard deviation for the data.

| Class limits | Frequency |
|--------------|-----------|
| 13–19        | 2         |
| 20–26        | 7         |
| 27–33        | 12        |
| 34–40        | 5         |
| 41–47        | 6         |
| 48–54        | 1         |
| 55–61        | 0         |
| 62–68        | 2         |

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- 34. Calories** The average number of calories in a regular-size bagel is 240. If the standard deviation is 38 calories, find the range in which at least 75% of the data will lie. Use Chebyshev's theorem.

# Exercises

- 34. Calories** The average number of calories in a regular-size bagel is 240. If the standard deviation is 38 calories, find the range in which at least 75% of the data will lie. Use Chebyshev's theorem.
- 35. Time Spent Online** Americans spend an average of 3 hours per day online. If the standard deviation is 32 minutes, find the range in which at least 88.89% of the data will lie. Use Chebyshev's theorem.
- 38. Trials to Learn a Maze** The average of the number of trials it took a sample of mice to learn to traverse a maze was 12. The standard deviation was 3. Using Chebyshev's theorem, find the minimum percentage of data values that will fall in the range of 4–20 trials.
- 39. Farm Sizes** The average farm in the United States in 2004 contained 443 acres. The standard deviation is 42 acres. Use Chebyshev's theorem to find the minimum percentage of data values that will fall in the range of 338–548 acres.



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# Thank You

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